

CONVOLUTIVE SEQUENCES, II: PARAMETRIZATIONS

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ABSTRACT. In recent work, the authors defined a sequence $(a_n)_{n \geq 0}$ to be m -convolutive exactly if

$$\sum_{n \geq 0} a_{mn} q^n = \left(\sum_{n \geq 0} a_n q^n \right)^m$$

for a specific positive integer m and provided proofs of the 2- and 3-convolutivity of a small number of sequences arising from primitive eta-products. Since the completion of that work, the authors have discovered many new instances of convolutive eta-products. The main focus of this work is to unify all but one of these instances in a parametric way.

1. INTRODUCTION

In the inaugural paper of this series, the authors introduced the concept of m -convolutivity for a sequence $(a_n)_{n \geq 0}$ if it satisfies the relation [2, Definition 1.1]

$$\sum_{n \geq 0} a_{mn} q^n = \left(\sum_{n \geq 0} a_n q^n \right)^m \quad (1.1)$$

with m a specific positive integer, and the generating series of a_n is also called m -convolutive. Meanwhile, the left-hand side of this defining relation is usually represented in terms of the *unitizing operator of degree m* ,

$$\mathbf{U}_m \left(\sum_{n \geq 0} a_n q^n \right) := \sum_{n \geq 0} a_{mn} q^n.$$

The main objective of [2] revolves around the case where the numbers a_n are generated by a primitive eta-product. As in [2, eq. (1.3)], for distinct positive integers m_j sorted in ascending order and integer exponents δ_j , we consider the *eta-product*

$$\prod_{j=1}^J (q^{m_j}; q^{m_j})_{\infty}^{\delta_j}, \quad (1.2)$$

where the standard q -Pochhammer symbol is given by

$$(a; q)_{\infty} := \prod_{k \geq 0} (1 - aq^k),$$

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while the shorthand notation

$$f_i := (q^i; q^i)_\infty$$

will be widely utilized throughout. We say the eta-product in (1.2) is *primitive* if the moduli $m_1, \dots, m_J$ have greatest common divisor 1. In [2], we proved the 2- and 3-convolutivity for a small list of sequences of numbers acting as the Fourier coefficients of primitive eta-products. All of these sequences can be found in the OEIS database [7].

Now a natural question to ask is — can we understand most of these convolutive relations, if not all, in a unified manner? In this direction, two of the authors [4] have already made progress in their study of partitions into odd parts with designated summands. In essence, their result can be rephrased in the following *parametric* way, and the equivalence to its original form in [4, p. 3, Theorem 1.3] (see Theorem 2.1) will be explained in Section 2.

Theorem 1.1 (Fu–Sellers [4], equivalent form). *Let X be independent of q . Then*

$$\mathbf{U}_2 \left(\frac{f_2^2}{f_1^2}(qX, q/X; q^2)_\infty \right) = \left(\frac{f_2^2}{f_1^2}(qX, q/X; q^2)_\infty \right)^2. \quad (1.3)$$

This parametric relation is of particular importance because it places four of the five 2-convolutive eta-products in our first paper within the same setting. By choosing X to be 1, -1 , $\omega := e^{\frac{2\pi i}{3}}$, $i := e^{\frac{\pi i}{2}}$, and $\zeta := e^{\frac{\pi i}{3}}$, respectively, we have

$$\begin{aligned} (q, q; q^2)_\infty &= \frac{f_1^2}{f_2^2}, & (-q, -q; q^2)_\infty &= \frac{f_2^4}{f_1^2 f_4^2}, \\ (\omega q, q/\omega; q^2)_\infty &= \frac{f_2 f_3}{f_1 f_6}, & (iq, q/i; q^2)_\infty &= \frac{f_4^2}{f_2 f_8}, \\ (\zeta q, q/\zeta; q^2)_\infty &= \frac{f_1 f_4 f_6^2}{f_2^2 f_3 f_{12}}. \end{aligned}$$

Then the eta-products in [2, Theorems 3.7, 3.6, 3.8, and 3.4], listed as the first four series in Table 1, are, respectively,

$$\begin{aligned} \frac{f_2^2}{f_1^2}(-q, -q; q^2)_\infty &= \frac{f_2^6}{f_1^4 f_4^2}, & \frac{f_2^2}{f_1^2}(\omega q, q/\omega; q^2)_\infty &= \frac{f_2^3 f_3}{f_1^3 f_6}, \\ \frac{f_2^2}{f_1^2}(iq, q/i; q^2)_\infty &= \frac{f_2 f_4^2}{f_1^2 f_8}, & \frac{f_2^2}{f_1^2}(\zeta q, q/\zeta; q^2)_\infty &= \frac{f_4 f_6^2}{f_1 f_3 f_{12}}. \end{aligned}$$

It should be noted that the search for convolutive sequences in our first paper [2] was conducted only on OEIS entries, and this was insufficient to detect all possible 2-convolutive eta-products for our parametrization purposes. Hereby, we execute a new search, this time directly for primitive eta-products. In doing so, we identify two more 2-convolutive examples, represented by the series (II.5) and (II.6) in Table 1.

Now the key observation is that

$$\begin{aligned} \frac{f_2 f_6}{f_1 f_3}(q, q^2, q, q^2; q^3)_\infty &= \frac{f_1 f_2 f_6}{f_3^3}, \\ \frac{f_2 f_6}{f_1 f_3}(-q, -q^2, -q, -q^2; q^3)_\infty &= \frac{f_2^3 f_3}{f_1^3 f_6}, \end{aligned}$$

TABLE 1. Primitive 2-convolutive eta-products

Key:			
§Newly discovered			
†Parametric, Thm. 1.1		‡Parametric, Thm. 1.2	
(II.1) [†]	$\frac{f_2^6}{f_1^4 f_4^2}$	(II.2) ^{†,‡}	$\frac{f_2^3 f_3}{f_1^3 f_6}$
(II.3) [†]	$\frac{f_2 f_4^2}{f_1^2 f_8}$	(II.4) ^{†,‡}	$\frac{f_4 f_6^2}{f_1 f_3 f_{12}}$
(II.5) ^{§,‡}	$\frac{f_1 f_2 f_6}{f_3^3}$	(II.6) ^{§,‡}	$\frac{f_2 f_3 f_6}{f_1^2 f_9}$
(II.7)	$\frac{f_6 f_{10}}{f_1 f_{15}}$		

$$\frac{f_2 f_6}{f_1 f_3}(\omega q, q^2/\omega, q/\omega, \omega q^2; q^3)_\infty = \frac{f_2 f_3 f_6}{f_1^2 f_9},$$

$$\frac{f_2 f_6}{f_1 f_3}(iq, q^2/i, q/i, iq^2; q^3)_\infty = \frac{f_4 f_6^2}{f_1 f_3 f_{12}}.$$

These are (II.5), (II.2), (II.6), and (II.4) in Table 1, respectively. Also, we find that

$$\frac{f_2 f_6}{f_1 f_3}(\zeta q, q^2/\zeta, q/\zeta, \zeta q^2; q^3)_\infty = \frac{f_6^3 f_9}{f_3^3 f_{18}}.$$

However, this relation does not produce any new information because for l and m coprime, if a series $S(q)$ is m -convolutive, then so is the series $S(q^l)$, and vice versa. In view of this fact, the primitive counterpart of the above eta-product, obtained by replacing q^3 with q , is exactly (II.2).

The previous discussions can be summarized as a second parametrization for 2-convolutive series, with the two newly discovered primitive eta-products (II.5) and (II.6) encoded.

Theorem 1.2. *Let X be independent of q . Then*

$$\mathbf{U}_2 \left(\frac{f_2 f_6}{f_1 f_3}(qX, q^2/X, q^2 X, q/X; q^3)_\infty \right) = \left(\frac{f_2 f_6}{f_1 f_3}(qX, q^2/X, q^2 X, q/X; q^3)_\infty \right)^2. \quad (1.4)$$

Remark 1.3. Both parametrizations (1.3) and (1.4) do not specialize to the last eta-product, (II.7), in Table 1, making this series the most mysterious. Such an eta-product will be called *sporadic*, as opposed to the *parametric* ones.

For 3-convolutive eta-products, one may wonder if a similar parametric generalization exists. However, in [2] only four 3-convolutive examples were recorded, making it less likely to find patterns. As such, we begin with a direct search for 3-convolutive eta-products, which suggests the primitive examples in Table 2, with the series (III.1) and (III.2) already proven in [2, Theorems 4.3 and 4.4] and the series (III.1') and (III.2') also justified in [2, Remark 4.1], while all others are newly discovered.

TABLE 2. Primitive 3-convolutive eta-products

Key: [§] Newly discovered			
$F(q)$		$F(-q)$	
(III.1)	$\frac{f_2 f_3^2}{f_1^2 f_6}$	(III.1')	$\frac{f_1^2 f_4^2 f_6^5}{f_2^5 f_3^2 f_{12}^2}$
(III.2)	$\frac{f_2 f_6^3}{f_1 f_3 f_4 f_{12}}$	(III.2')	$\frac{f_1 f_3}{f_2^2}$
(III.3) [§]	$\frac{f_2^7 f_6}{f_1^4 f_4^4}$	(III.3') [§]	$\frac{f_1^4 f_6}{f_2^5}$
(III.4) [§]	$\frac{f_2^4 f_3}{f_1^3 f_4^2}$	(III.4') [§]	$\frac{f_1^3 f_4 f_6^3}{f_2^5 f_3 f_{12}}$
(III.5) [§]	$\frac{f_3 f_4^2}{f_1 f_2 f_8}$	(III.5') [§]	$\frac{f_1 f_4^3 f_6^3}{f_2^4 f_3 f_8 f_{12}}$
(III.6) [§]	$\frac{f_2^2 f_6}{f_1^2 f_8}$	(III.6') [§]	$\frac{f_1^2 f_4^2 f_6}{f_2^4 f_8}$
(III.7) [§]	$\frac{f_5 f_6}{f_1 f_{10}}$	(III.7') [§]	$\frac{f_1 f_4 f_6 f_{10}^2}{f_2^3 f_5 f_{20}}$

We remark that Table 2 should be considered in the following way. Given a sequence $(a_n)_{n \geq 0}$, its *dual* sequence $(a'_n)_{n \geq 0}$ is defined by the relation $a'_n := (-1)^n a_n$. As noted in [2, Remark 4.1], if a sequence is 3-convolutive, then so is its dual, and vice versa. In this sense, the series listed in Table 2 can be naturally paired, and once one series in such a pair is proven to be 3-convolutive, then so is the other. More precisely, the series labeled by a prime can be obtained by replacing q with $-q$ in the corresponding series; such equalities can be shown by the simple q -series relation that

$$(-q; -q)_\infty = \frac{f_2^3}{f_1 f_4}.$$

In view of the first six pairs in Table 2, the following pattern occurs:

$$\begin{aligned}
\frac{f_2 f_3^2}{f_1^2 f_6} &= \frac{f_6}{f_2}(\omega q, q/\omega, \omega q, q/\omega; q^2)_\infty, & \frac{f_1^2 f_4^2 f_6^5}{f_2^5 f_3^2 f_{12}^2} &= \frac{f_6}{f_2}(\zeta q, q/\zeta, \zeta q, q/\zeta; q^2)_\infty, \\
\frac{f_2 f_6^3}{f_1 f_3 f_4 f_{12}} &= \frac{f_6}{f_2}(-q, -q, \zeta q, q/\zeta; q^2)_\infty, & \frac{f_1 f_3}{f_2^2} &= \frac{f_6}{f_2}(q, q, \omega q, q/\omega; q^2)_\infty, \\
\frac{f_2^7 f_6}{f_1^4 f_4^4} &= \frac{f_6}{f_2}(-q, -q, -q, -q; q^2)_\infty, & \frac{f_1^4 f_6}{f_2^5} &= \frac{f_6}{f_2}(q, q, q, q; q^2)_\infty, \\
\frac{f_2^4 f_3}{f_1^3 f_4^2} &= \frac{f_6}{f_2}(-q, -q, \omega q, q/\omega; q^2)_\infty, & \frac{f_1^3 f_4 f_6^3}{f_2^5 f_3 f_{12}} &= \frac{f_6}{f_2}(q, q, \zeta q, q/\zeta; q^2)_\infty,
\end{aligned}$$

$$\begin{aligned}\frac{f_3 f_4^2}{f_1 f_2 f_8} &= \frac{f_6}{f_2}(\omega q, q/\omega, iq, q/i; q^2)_\infty, & \frac{f_1 f_4^3 f_6^3}{f_2^4 f_3 f_8 f_{12}} &= \frac{f_6}{f_2}(iq, q/i, \zeta q, q/\zeta; q^2)_\infty, \\ \frac{f_2^2 f_6}{f_1^2 f_8} &= \frac{f_6}{f_2}(-q, -q, iq, q/i; q^2)_\infty, & \frac{f_1^2 f_4^2 f_6}{f_2^4 f_8} &= \frac{f_6}{f_2}(q, q, iq, q/i; q^2)_\infty.\end{aligned}$$

Moreover, letting $\rho := e^{\frac{\pi i}{5}}$, we find that the same pattern also works for the last pair:

$$\frac{f_5 f_6}{f_1 f_{10}} = \frac{f_6}{f_2}(\rho^2 q, q/\rho^2, \rho^4 q, q/\rho^4; q^2)_\infty, \quad \frac{f_1 f_4 f_6 f_{10}^2}{f_2^3 f_5 f_{20}} = \frac{f_6}{f_2}(\rho q, q/\rho, \rho^3 q, q/\rho^3; q^2)_\infty.$$

Hence, we are led to consider the biparametric 3-convolutive relation in the next theorem, which surprisingly encodes *all* fourteen primitive eta-products in Table 2.

Theorem 1.4. *Let X and Y be independent of q . Then*

$$\mathbf{U}_3 \left(\frac{f_6}{f_2}(qX, q/X, qY, q/Y; q^2)_\infty \right) = \left(\frac{f_6}{f_2}(qX, q/X, qY, q/Y; q^2)_\infty \right)^3. \quad (1.5)$$

Notably, if we want the series

$$\frac{f_6}{f_2}(qX, q/X, qY, q/Y; q^2)_\infty$$

in Theorem 1.4 to be an eta-product, while $X, Y \in \{\rho, \rho^2, \rho^3, \rho^4\}$, the only possibilities are given by the last pair in Table 2. Meanwhile, if we require $X, Y \in \{1, -1, \omega, i, \zeta\}$, then the series specializes to twelve different primitive eta-products, all appearing among the first six pairs in the same table. There are also three nonprimitive cases:

$$\begin{aligned}\frac{f_6}{f_2}(q, q, -q, -q; q^2)_\infty &= \frac{f_2 f_6}{f_4^2}, & \frac{f_6}{f_2}(\omega q, q/\omega, \zeta q, q/\zeta; q^2)_\infty &= \frac{f_4 f_6^2}{f_2^2 f_{12}}, \\ \frac{f_6}{f_2}(iq, q/i, iq, q/i; q^2)_\infty &= \frac{f_4^4 f_6}{f_2^3 f_8^2},\end{aligned}$$

and their primitive counterparts, obtained by replacing q^2 with q throughout each eta-product, are still among the aforementioned twelve members.

Outline of the paper. In Section 2, we review the combinatorial motivation described in [4] for our parametric 2-convolutivity in Theorem 1.1 and demonstrate the equivalence between (1.3) and the combinatorial relation (2.3). In addition, we connect a uniparametric specialization of the 3-convolutive relation in Theorem 1.4 to a signed counting for partitions with designated summands in a manner akin to that in [4]. Following this, in Section 3, we provide an alternative proof of Theorem 1.1 using a theta-dissection technique, while in Sections 4 and 5, we establish Theorems 1.2 and 1.4, respectively, based on an analogous but more delicate analysis. Finally, we close in Section 6 with some comments and questions for future study.

2. COMBINATORIAL MOTIVATION

Andrews, Lewis, and Lovejoy [1] introduced *partitions with designated summands* as partitions such that exactly one part of each size in the partition is marked. For example, $3' + 3 + 2 + 2' + 2 + 1'$ is a partition of 13 with designated summands in

which we mark the first part of size 3, the second part of size 2, and the only part of size 1.

Denote by $\mathcal{PD}$ the set of partitions with designated summands, and further by $\mathcal{PD}\mathcal{O}$ the set of partitions with designated summands wherein all parts are odd. An important discovery of Andrews, Lewis, and Lovejoy is the generating function identity [1, p. 52, eq. (1.6)]

$$\sum_{\lambda \in \mathcal{PD}\mathcal{O}} q^{|\lambda|} = \frac{f_4 f_6^2}{f_1 f_3 f_{12}},$$

where $|\lambda|$ is the *weight* of the partition λ , namely, the sum of all its parts. In particular, the eta-product on the right-hand side of the above satisfies the 2-convolutive relation [1, p. 63, Theorem 21]

$$\mathbf{U}_2 \left(\frac{f_4 f_6^2}{f_1 f_3 f_{12}} \right) = \left(\frac{f_4 f_6^2}{f_1 f_3 f_{12}} \right)^2, \quad (2.1)$$

thereby yielding the following combinatorial result:

$$\mathbf{U}_2 \left(\sum_{\lambda \in \mathcal{PD}\mathcal{O}} q^{|\lambda|} \right) = \left(\sum_{\lambda \in \mathcal{PD}\mathcal{O}} q^{|\lambda|} \right)^2. \quad (2.2)$$

For a partition λ with designated summands, let $\text{div}(\lambda)$ be the *diversity* of λ , that is, the number of different part sizes in λ . In [4, p. 3, Theorem 1.3], two of the authors provided a uniparametric generalization of (2.2).

Theorem 2.1 (Fu–Sellers [4], original form). *Let x be independent of q . Then*

$$\mathbf{U}_2 \left(\sum_{\lambda \in \mathcal{PD}\mathcal{O}} x^{\text{div}(\lambda)} q^{|\lambda|} \right) = \left(\sum_{\lambda \in \mathcal{PD}\mathcal{O}} x^{\text{div}(\lambda)} q^{|\lambda|} \right)^2. \quad (2.3)$$

Now we show why this combinatorial relation is equivalent to our analytic parametrization in (1.3).

Proof of the equivalence between (1.3) and (2.3). By definition,

$$\begin{aligned} \sum_{\lambda \in \mathcal{PD}\mathcal{O}} x^{\text{div}(\lambda)} q^{|\lambda|} &= \prod_{i \geq 1} (1 + xq^{2i-1} + 2xq^{2(2i-1)} + 3xq^{3(2i-1)} + \dots) \\ &= \frac{f_2^2}{f_1^2} \prod_{i \geq 1} (1 + (x-2)q^{2i-1} + q^{4i-2}). \end{aligned}$$

Making the change of variables

$$X = X(x) := \frac{2 - x + \sqrt{x^2 - 4x}}{2}, \quad (2.4)$$

we have

$$\sum_{\lambda \in \mathcal{PD}\mathcal{O}} x^{\text{div}(\lambda)} q^{|\lambda|} = \frac{f_2^2}{f_1^2} (qX, q/X; q^2)_\infty. \quad (2.5)$$

Since x is independent of q , so is X . The equivalence between (1.3) and (2.3) is now clear. $\square$

Noting the appearance of the factors $(qX, q/X; q^2)_\infty$ and $(qY, q/Y; q^2)_\infty$ in the 3-convolutive relation (1.5), our next question is whether it has a combinatorial version in analogy with (2.3). Although it is possible to interpret these two factors in terms of pairs of partitions in $\mathcal{PD}\mathcal{O}$, this explanation is not as neat as we wish. In this sense, we have to sacrifice the degree of freedom by considering a uniparametric specialization.

For the moment, let $\text{len}_e(\lambda)$ and $\text{div}_o(\lambda)$ denote the total number of even parts and the number of different odd part sizes in $\lambda \in \mathcal{PD}$, respectively. We have the following *signed* counting for partitions with designated summands:

$$\begin{aligned} & \sum_{\lambda \in \mathcal{PD}} (-1)^{\text{len}_e(\lambda)} x^{\text{div}_o(\lambda)} q^{|\lambda|} \\ &= \prod_{i \geq 1} (1 + xq^{2i-1} + 2xq^{2(2i-1)} + \dots) (1 - q^{2i} + 2q^{4i} - \dots) \\ &= \frac{f_2^3 f_6}{f_1^2 f_4^2} \prod_{i \geq 1} (1 + (x-2)q^{2i-1} + q^{4i-2}). \end{aligned}$$

Using the same change of variables as in (2.4),

$$X = X(x) := \frac{2 - x + \sqrt{x^2 - 4x}}{2},$$

we derive the generating function identity

$$\sum_{\lambda \in \mathcal{PD}} (-1)^{\text{len}_e(\lambda)} x^{\text{div}_o(\lambda)} q^{|\lambda|} = \frac{f_2^3 f_6}{f_1^2 f_4^2} (qX, q/X; q^2)_\infty. \quad (2.6)$$

Now the key observation is that the right-hand of (2.6) is exactly the 3-convolutive series in (1.5) with $Y = -1$. That is,

$$\frac{f_2^3 f_6}{f_1^2 f_4^2} (qX, q/X; q^2)_\infty = \frac{f_6}{f_2} (qX, q/X, -q, -q; q^2)_\infty.$$

Therefore, we arrive at a neat 3-convolutive analog to (2.3) in light of Theorem 1.4.

Corollary 2.2. *Let x be independent of q . Then*

$$\mathbf{U}_3 \left(\sum_{\lambda \in \mathcal{PD}} (-1)^{\text{len}_e(\lambda)} x^{\text{div}_o(\lambda)} q^{|\lambda|} \right) = \left(\sum_{\lambda \in \mathcal{PD}} (-1)^{\text{len}_e(\lambda)} x^{\text{div}_o(\lambda)} q^{|\lambda|} \right)^3. \quad (2.7)$$

3. FIRST 2-CONVOLUTIVE PARAMETRIZATION, A THETA-DISSECTION PROOF

It is notable that, in [4], the combinatorial relation (2.3) was demonstrated by the theory of Chebyshev polynomials. In this section, we provide an alternative, and in some sense, more intrinsic proof by a theta-dissection technique applied to the series in (1.3).

To begin with, we share the following simple yet useful criterion for proving m -convolutive relations.

Lemma 3.1. *Fix $m \geq 2$. Assume $A(q), B(q) \in \mathbb{Z}[[q]]$ with $B(q)$ not identical to zero. Then the product $A(q)B(q^m)$ is m -convolutive if and only if*

$$\mathbf{U}_m(A(q)) = A(q)^m \cdot \frac{B(q^m)^m}{B(q)}. \quad (3.1)$$

Proof. According to the defining relation (1.1), we see that $A(q)B(q^m)$ is m -convolutive if and only if

$$(A(q)B(q^m))^m = \mathbf{U}_m(A(q)B(q^m)) = \mathbf{U}_m(A(q)) \cdot B(q),$$

which matches (3.1) after rearranging the factors. $\square$

Now we are ready to show Theorem 1.1.

Proof of Theorem 1.1. In Lemma 3.1, we fix $m = 2$ and choose

$$A(q) = \frac{1}{f_1^2}(qX, q/X, q^2; q^2)_\infty, \quad B(q) = f_1.$$

Then it remains to show

$$\mathbf{U}_2(A(q)) = \frac{f_2^2}{f_1^5}(qX, q/X, q^2; q^2)_\infty^2.$$

Recall from [5, p. 14, eq. (1.9.4)] with q replaced by $-q$ that

$$\frac{1}{f_1^2} = \frac{f_8^5}{f_2^5 f_{16}^2} + 2q \frac{f_4^2 f_{16}^2}{f_2^5 f_8}.$$

Meanwhile, by the Jacobi triple product identity [5, p. 1, eq. (1.1.1)], we have

$$\begin{aligned} (qX, q/X, q^2; q^2)_\infty &= \sum_{m=-\infty}^{\infty} (-1)^m X^m q^{m^2} \\ &= \sum_{m=-\infty}^{\infty} X^{2m} q^{(2m)^2} - \sum_{m=-\infty}^{\infty} X^{2m+1} q^{(2m+1)^2} \\ &= (-q^4 X^2, -q^4/X^2, q^8; q^8)_\infty - qX(-q^8 X^2, -1/X^2, q^8; q^8)_\infty. \end{aligned}$$

Thus,

$$\begin{aligned} \mathbf{U}_2(A(q)) &= \frac{f_2^2}{f_1^5} \left(\frac{f_4^5}{f_2^2 f_8^2} (-q^2 X^2, -q^2/X^2, q^4; q^4)_\infty \right. \\ &\quad \left. - 2qX \frac{f_8^2}{f_4} (-q^4 X^2, -1/X^2, q^4; q^4)_\infty \right), \end{aligned}$$

so that it is sufficient to show that

$$\begin{aligned} (qX, q/X, q^2; q^2)_\infty^2 &= \frac{f_4^5}{f_2^2 f_8^2} (-q^2 X^2, -q^2/X^2, q^4; q^4)_\infty \\ &\quad - 2qX \frac{f_8^2}{f_4} (-q^4 X^2, -1/X^2, q^4; q^4)_\infty. \end{aligned}$$

Finally, we note that

$$\begin{aligned} (qX, q/X, q^2; q^2)_\infty^2 &= \sum_{n_1, n_2=-\infty}^{\infty} (-1)^{n_1-n_2} X^{n_1+n_2} q^{n_1^2+n_2^2} \\ &= \sum_{\substack{m_1, m_2=-\infty \\ m_1 \equiv m_2 \pmod{2}}}^{\infty} (-1)^{m_1} X^{m_2} q^{\frac{1}{2}(m_1^2+m_2^2)} \end{aligned}$$

$$\begin{aligned}
&= \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{2(l_1^2 + l_2^2)} - qX \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{2(l_1^2 + l_2^2) + 2(l_1 + l_2)} \\
&= (-q^2, -q^2, q^4; q^4)_{\infty} (-q^2 X^2, -q^2/X^2, q^4; q^4)_{\infty} \\
&\quad - qX(-q^4, -1, q^4; q^4)_{\infty} (-q^4 X^2, -1/X^2, q^4; q^4)_{\infty},
\end{aligned}$$

as desired. $\square$

4. SECOND 2-CONVOLUTIVE PARAMETRIZATION

Using the same theta-dissection technique but with a more intricate analysis, we can provide an analogous proof for Theorem 1.2. Let us start by rewriting the series in (1.4) as

$$\begin{aligned}
\frac{f_2 f_6}{f_1 f_3} (qX, q^2/X, q^2 X, q/X; q^3)_{\infty} &= \frac{1}{f_1 f_3} (qX, q^5/X, q^6; q^6)_{\infty} (q^5 X, q/X, q^6; q^6)_{\infty} \\
&\quad \times \frac{f_2}{f_6} (q^2 X, q^4/X, q^4 X, q^2/X; q^6)_{\infty}.
\end{aligned}$$

Now in Lemma 3.1, we fix $m = 2$ and choose

$$A(q) = \frac{1}{f_1 f_3} (qX, q^5/X, q^6; q^6)_{\infty} (q^5 X, q/X, q^6; q^6)_{\infty}$$

and

$$B(q) = \frac{f_1}{f_3} (qX, q^2/X, q^2 X, q/X; q^3)_{\infty}.$$

Then it remains to show

$$\mathbf{U}_2(A(q)) = \frac{f_2^2 f_6^2}{f_1^3 f_3^3} (qX, q^2/X, q^3; q^3)_{\infty} (q^2 X, q/X, q^3; q^3)_{\infty}.$$

Recall from [5, p. 218, eq. (25.1.7)] that

$$\frac{1}{f_1 f_3} = \frac{f_8^2 f_{12}^5}{f_2^2 f_4 f_6^4 f_{24}^2} + q \frac{f_4^5 f_{24}^2}{f_2^4 f_6^2 f_8^2 f_{12}}.$$

Also, by the Jacobi triple product identity,

$$\begin{aligned}
&(qX, q^5/X, q^6; q^6)_{\infty} (q^5 X, q/X, q^6; q^6)_{\infty} \\
&= \sum_{n_1, n_2 = -\infty}^{\infty} (-1)^{n_1 + n_2} X^{n_2 - n_1} q^{3(n_1^2 + n_2^2) + 2(n_1 + n_2)} \\
&= \sum_{\substack{m_1, m_2 = -\infty \\ m_1 \equiv m_2 \pmod{2}}}^{\infty} (-1)^{m_1} X^{m_2} q^{\frac{3}{2}(m_1^2 + m_2^2) + 2m_1} \\
&= \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{6l_1^2 + 6l_2^2 + 4l_1} - qX \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{6l_1^2 + 6l_2^2 - 2l_1 + 6l_2} \\
&= (-q^2, -q^{10}, q^{12}; q^{12})_{\infty} (-q^6 X^2, -q^6/X^2, q^{12}; q^{12})_{\infty} \\
&\quad - qX(-q^4, -q^8, q^{12}; q^{12})_{\infty} (-q^{12} X^2, -1/X^2, q^{12}; q^{12})_{\infty} \\
&= \frac{f_4^2 f_6 f_{24}}{f_2 f_8 f_{12}} (-q^6 X^2, -q^6/X^2, q^{12}; q^{12})_{\infty} - qX \frac{f_8 f_{12}^2}{f_4 f_{24}} (-q^{12} X^2, -1/X^2, q^{12}; q^{12})_{\infty}.
\end{aligned}$$

It follows that

$$\begin{aligned} \mathbf{U}_2(A(q)) &= \frac{f_2 f_4 f_6^4}{f_1^3 f_3^3 f_{12}} (-q^3 X^2, -q^3/X^2, q^6; q^6)_\infty \\ &\quad - qX \frac{f_2^4 f_6 f_{12}}{f_1^4 f_3^2 f_4} (-q^6 X^2, -1/X^2, q^6; q^6)_\infty. \end{aligned}$$

Finally, we note that

$$\begin{aligned} &(qX, q^2/X, q^3; q^3)_\infty (q^2 X, q/X, q^3; q^3)_\infty \\ &= \sum_{n_1, n_2 = -\infty}^{\infty} (-1)^{n_1+n_2} X^{n_2-n_1} q^{\frac{3}{2}(n_1^2+n_2^2)+\frac{1}{2}(n_1+n_2)} \\ &= \sum_{\substack{m_1, m_2 = -\infty \\ m_1 \equiv m_2 \pmod{2}}}^{\infty} (-1)^{m_1} X^{m_2} q^{\frac{3}{4}(m_1^2+m_2^2)+\frac{1}{2}m_1} \\ &= \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{3l_1^2+3l_2^2+l_1} - qX \sum_{l_1, l_2 = -\infty}^{\infty} X^{2l_2} q^{3l_1^2+3l_2^2-2l_1+3l_2} \\ &= (-q^2, -q^4, q^6; q^6)_\infty (-q^3 X^2, -q^3/X^2, q^6; q^6)_\infty \\ &\quad - qX (-q, -q^5, q^6; q^6)_\infty (-q^6 X^2, -1/X^2, q^6; q^6)_\infty \\ &= \frac{f_4 f_6^2}{f_2 f_{12}} (-q^3 X^2, -q^3/X^2, q^6; q^6)_\infty - qX \frac{f_2^2 f_3 f_{12}}{f_1 f_4 f_6} (-q^6 X^2, -1/X^2, q^6; q^6)_\infty, \end{aligned}$$

so that

$$\begin{aligned} &\frac{f_2^2 f_6^2}{f_1^3 f_3^3} (qX, q^2/X, q^3; q^3)_\infty (q^2 X, q/X, q^3; q^3)_\infty \\ &= \frac{f_2 f_4 f_6^4}{f_1^3 f_3^3 f_{12}} (-q^3 X^2, -q^3/X^2, q^6; q^6)_\infty - qX \frac{f_2^4 f_6 f_{12}}{f_1^4 f_3^2 f_4} (-q^6 X^2, -1/X^2, q^6; q^6)_\infty. \end{aligned}$$

This matches the expression for $\mathbf{U}_2(A(q))$ we obtained earlier, thereby concluding the proof.

5. 3-CONVOLUTIVE PARAMETRIZATION

To prove Theorem 1.4, we fix $m = 3$ in Lemma 3.1 and choose

$$A(q) = \frac{1}{f_2^3} (qX, q/X, q^2; q^2)_\infty (qY, q/Y, q^2; q^2)_\infty, \quad B(q) = f_2.$$

It remains to show

$$\mathbf{U}_3(A(q)) = \frac{f_6^3}{f_2^{10}} (qX, q/X, q^2; q^2)_\infty^3 (qY, q/Y, q^2; q^2)_\infty^3.$$

Recall from [5, p. 184, eq. (21.3.7)] that

$$f_1^3 = a(q^3) f_3 - 3q f_9^3,$$

where according to [5, p. 179, eqs. (21.1.1) and (21.1.2)],

$$a(q) := 1 + 6 \sum_{n=0}^{\infty} \left(\frac{q^{3n+1}}{1-q^{3n+1}} - \frac{q^{3n+2}}{1-q^{3n+2}} \right).$$

Now replacing q with ωq and $\omega^2 q$ in the above where $\omega := e^{\frac{2\pi i}{3}}$, and then multiplying the two results, we have

$$\frac{1}{f_1^3} = a(q^3)^2 \frac{f_9^3}{f_3^{10}} + 3qa(q^3) \frac{f_9^6}{f_3^{11}} + 9q^2 \frac{f_9^9}{f_3^{12}}.$$

Meanwhile, by the Jacobi triple product identity, this time with a 3-dissection, we have

$$\begin{aligned} & (qX, q/X, q^2; q^2)_\infty \\ &= \sum_{m=-\infty}^{\infty} (-1)^m X^m q^{m^2} \\ &= \sum_{m=-\infty}^{\infty} (-1)^m X^{3m} q^{(3m)^2} - \sum_{m=-\infty}^{\infty} (-1)^m (X^{3m+1} + X^{-3m-1}) q^{(3m+1)^2} \\ &= F_1(X, q^3) - qX F_2(X, q^3) - qX^{-1} F_3(X, q^3), \end{aligned}$$

where

$$\begin{aligned} F_1(X, q) &:= (q^3 X^3, q^3/X^3, q^6; q^6)_\infty, \\ F_2(X, q) &:= (q^5 X^3, q/X^3, q^6; q^6)_\infty, \\ F_3(X, q) &:= (qX^3, q^5/X^3, q^6; q^6)_\infty. \end{aligned}$$

Thus,

$$\begin{aligned} \mathbf{U}_3(A(q)) &= a(q^2)^2 F_1(X, q) F_1(Y, q) \frac{f_6^3}{f_2^{10}} \\ &\quad - 3qa(q^2) F_1(X, q) (Y F_2(Y, q) + Y^{-1} F_3(Y, q)) \frac{f_6^6}{f_2^{11}} \\ &\quad - 3qa(q^2) F_1(Y, q) (X F_2(X, q) + X^{-1} F_3(X, q)) \frac{f_6^6}{f_2^{11}} \\ &\quad + 9q^2 (X F_2(X, q) + X^{-1} F_3(X, q)) (Y F_2(Y, q) + Y^{-1} F_3(Y, q)) \frac{f_6^9}{f_2^{12}}. \end{aligned}$$

Finally, we need [5, p. 180, eq. (21.2.4)]

$$(qX, q/X, q^2; q^2)_\infty^3 = a(q^2) F_1(X, q) - 3q \frac{f_6^3}{f_2} (X F_2(X, q) + X^{-1} F_3(X, q)).$$

It follows that

$$\begin{aligned} & \frac{f_6^3}{f_2^{10}} (qX, q/X, q^2; q^2)_\infty^3 (qY, q/Y, q^2; q^2)_\infty^3 \\ &= \frac{f_6^3}{f_2^{10}} \left(a(q^2) F_1(X, q) - 3q \frac{f_6^3}{f_2} (X F_2(X, q) + X^{-1} F_3(X, q)) \right) \\ &\quad \times \left(a(q^2) F_1(Y, q) - 3q \frac{f_6^3}{f_2} (Y F_2(Y, q) + Y^{-1} F_3(Y, q)) \right), \end{aligned}$$

which is exactly the same as the previous expression for $\mathbf{U}_3(A(q))$ after expanding the product. The proof is therefore complete.

6. CONCLUSION

We close this work with three sets of comments and questions.

First, an important problem not yet explored is the possibility of unifying the two 2-convolutive parametrizations (1.3) and (1.4) in a biparametric way similar to that for (1.5). Ideally, the related series should take the form

$$P(q)(qX, q/X; q^2)_\infty (qY, q^2/Y, q^2Y, q/Y; q^3)_\infty,$$

where the prefactor $P(q)$ is an eta-product. For the moment, specializing $(qX, q/X; q^2)_\infty$ to an eta-product, which may have the following options

$$\frac{f_1^2}{f_2^2}, \quad \frac{f_2^4}{f_1^2 f_4^2}, \quad \frac{f_2 f_3}{f_1 f_6}, \quad \frac{f_4^2}{f_2 f_8}, \quad \frac{f_1 f_4 f_6^2}{f_2^2 f_3 f_{12}},$$

then its product with $P(q)$ should be $\frac{f_2 f_6}{f_1 f_3}$, the prefactor in (1.4). Thus, $P(q)$ is one of

$$\frac{f_2^3 f_6}{f_1^3 f_3}, \quad \frac{f_1 f_4^2 f_6}{f_2^3 f_3}, \quad \frac{f_6^2}{f_3^2}, \quad \frac{f_2^2 f_6 f_8}{f_1 f_3 f_4^2}, \quad \frac{f_2^3 f_{12}}{f_1^2 f_4 f_6}.$$

Similarly, we may specialize $(qY, q^2/Y, q^2Y, q/Y; q^3)_\infty$ as

$$\frac{f_1^2}{f_3^2}, \quad \frac{f_2^2 f_3^2}{f_1^2 f_6^2}, \quad \frac{f_3^2}{f_1 f_9}, \quad \frac{f_4 f_6}{f_2 f_{12}}, \quad \frac{f_1 f_6^2 f_9}{f_2 f_3^2 f_{18}},$$

and multiplying it by $P(q)$ should give $\frac{f_2^2}{f_1^2}$, the prefactor in (1.3). Then the choices of $P(q)$ in this case include

$$\frac{f_2^2 f_3^2}{f_1^4}, \quad \frac{f_6^2}{f_3^2}, \quad \frac{f_2^2 f_9}{f_1 f_3^2}, \quad \frac{f_2^3 f_{12}}{f_1^2 f_4 f_6}, \quad \frac{f_2^3 f_3^2 f_{18}}{f_1^3 f_6^2 f_9}.$$

Now the overlapping expressions for $P(q)$ are

$$\frac{f_6^2}{f_3^2}, \quad \frac{f_2^3 f_{12}}{f_1^2 f_4 f_6}.$$

However, a direct verification reveals that neither of these options produces the desired 2-convolutivity.

Second, since all other 2- and 3-convolutive primitive eta-products in this work can be parametrized, it is curious to ask if the series (II.7) in Table 1 is *truly* sporadic, or if there is a missing parametrization associated with it. In addition, the parametric series in Theorems 1.1, 1.2, and 1.4 reduce to a primitive eta-product only for a small selection of parameters, all identified in Tables 1 and 2. It is still unclear if one can construct an infinite family of 2- or 3-convolutive sequences whose generating functions are primitive eta-products, or on the opposite side, if the list of primitive 2- and 3-convolutive eta-products is finite.

Third, in a separate project [3], we construct a bijective proof for the basic combinatorial relation (2.2). However, when the contribution of the “div” statistic is inserted, the bijectivity for (2.3) remains open. For the 3-convolutive case, it is an easy exercise, which will be left to the motivated reader, to show that the coefficient sequences represented in the left-hand column of Table 2 are all nonnegative. This fact suggests that the series in the left-hand column should be the counting functions for certain partition sets, thereby shedding light on potential combinatorial proofs of the 3-convolutivity. In this direction, we have witnessed an interesting paper

by Liu and Tian [6] on a combinatorial treatment of the series (III.1). For a more general consideration, the parametric series in (1.5) with X and Y replaced by $-X$ and $-Y$, namely,

$$\frac{f_6}{f_2}(-qX, -q/X, -qY, -q/Y; q^2)_\infty,$$

belongs to $\mathbb{N}[X, X^{-1}, Y, Y^{-1}][[q]]$, so it should be the counting function for certain partition tuples. It would be intriguing to see a bijective proof of (1.5) along this line. As we have remarked in Section 2, such a partition tuple interpretation might not be as neat as we would like to see. Then a combinatorial study of the neat relation (2.7), which corresponds to the $Y = -1$ case of (1.5), is also of great interest.

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